

# SYDNEY TECHNICAL HIGH SCHOOL



## EXTENSION 1

# MATHEMATICS

## PRELIMINARY ASSESSMENT TASK 2

JULY 2012

**Time Allowed:** 70 minutes

### Instructions:

- Write using blue or black pen.
- Approved calculators may be used.
- Attempt all questions.
- All necessary working must be shown. Marks may not be awarded for careless or badly arranged work.
- Marks indicated are a guide only and may be varied if necessary.
- **PLEASE START EACH NEW QUESTION ON A NEW PAGE.**

**Name:** \_\_\_\_\_

**Teacher:** \_\_\_\_\_

| Q1  | Q2  | Q3  | Q4  | Q5  | Q6  | TOTAL |
|-----|-----|-----|-----|-----|-----|-------|
| /10 | /10 | /10 | /10 | /10 | /10 | /60   |

**Question 1**

( 10 marks )

- a) What is the value of  $\tan\left(\frac{-3\pi}{4}\right)$  ? 1
- b) Find the exact value of  $\cos 105^\circ$  2
- c) Find the coordinates of the point which divides the interval AB,  
with A( 1, 4 ) and B( 5, 2 ), externally in the ratio 1 : 3. 2
- d) Differentiate
- i.  $\frac{4}{\sqrt{x}}$  1
- ii.  $(x^2 - 1)^{10}$  2
- iii.  $\frac{x}{x+1}$  2

**Question 2** START A NEW PAGE

( 10 marks )

- a) If  $\sin \alpha = \frac{3}{4}$ ,  $0 < \alpha < 90^\circ$  and  $\sin \beta = \frac{2}{3}$ ,  $90^\circ < \beta < 180^\circ$ ,  
find the exact values of :
- i.  $\tan 2\alpha$  2
- ii.  $\sin(\alpha - \beta)$  2
- b) Find  $\frac{dy}{dx}$  if  $y = 2x(x + 4)^8$  2
- c) Solve the equation  $\sin 2x = \tan x$  for  $0 \leq x \leq \pi$  4

**Question 3**    START A NEW PAGE

( 10 MARKS )

a) If  $\alpha$  and  $\beta$  are the roots of  $3x^2 - 4x + 8 = 0$ , find the value of :

i.  $\alpha\beta$  1

ii.  $(\alpha - \beta)^2$  2

b)

i. Show that  $\frac{1 + \cos 2A}{\sin 2A} = \cot A$  2

ii. Hence, find the exact value of  $\cot 15^\circ$  1

c) The curve  $y = ax^3 + bx$  cuts the  $x$  axis at  $x = 1$  and the gradient of the tangent at this point is 4.

i. Find the values of  $a$  and  $b$  2

ii. Find the acute angle between the tangent at  $x = 1$  and the line  $4x - 3y + 2 = 0$  2

**Question 4**    START A NEW PAGE

( 10 MARKS )

a) For the function  $y = \frac{1}{1+x^2}$

i. Find the derivative 2

ii. Determine the equation of the normal at the point where

$y = \frac{1}{2}$  and  $x < 0$  3

iii. Where does this normal cut the  $x$  axis? 1

b) Given  $\tan \frac{\theta}{2} = t$

i. Express  $\sec \theta + \tan \theta$  in terms of  $t$

( express your answer in its simplest form ) 2

ii. Hence, or otherwise, solve  $\sec \theta + \tan \theta = 2$ , correct to nearest degree for  $0^\circ < \theta < 90^\circ$  2

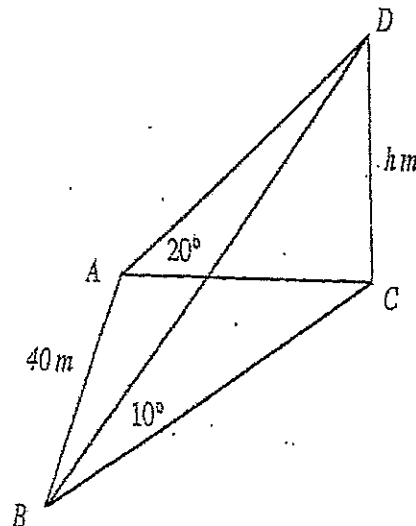
a) Show that  $(\sin A - \cos A)^2 = 1 - \sin 2A$  2

b)

i. Express  $\cos \theta - \sqrt{3} \sin \theta$  in the form  $R \cos(\theta + \alpha)$  for  $R > 0$  2

ii. Hence, or otherwise, solve  $\cos \theta - \sqrt{3} \sin \theta = 1$  for  $0 \leq \theta \leq 2\pi$  2

- c) A vertical flagpole  $CD$  of height  $h$  metres, stands with its base  $C$  on horizontal ground.  $A$  is a point on the ground due West of  $C$  and  $B$  is a point on the ground 40 metres due South of  $A$ . From  $A$  and  $B$  the angles of elevation of the top  $D$  of the flagpole are  $20^\circ$  and  $10^\circ$  respectively.



- i. Use triangle  $ADC$  to show that  $AC = h \tan 70^\circ$  1
- ii. Write a similar expression for  $BC$  1
- iii. Find the height of the flagpole correct to the nearest metre. 2

- a) For what value ( or values ) of  $k$ , will the quadratic equation  
 $(k + 4)x^2 - 3kx - 4(k - 2) = 0$  have two roots which differ by 1.  
( hint let the roots be  $\alpha$  and  $\alpha + 1$  ) 4
- b)
- i. State the conditions for the quadratic expression  $ax^2 + bx + c$   
to be negative definite. 1
- ii. Hence, or otherwise, show that the expression  
 $(k^2 + k)x^2 - (2k - 6)x + 2$  can never be negative definite 3
- iii. Find the range of values of  $k$  for which the expression is positive definite 2

***End of examination***

### Question 1

a)  $\tan\left(-\frac{3\pi}{4}\right)$

$= \tan(-135^\circ)$

$= 1 \quad \checkmark$

b)  $\cos 105^\circ$

$\cos(60^\circ + 45^\circ) \quad \checkmark$

$= \cos 60^\circ \cos 45^\circ - \sin 60^\circ \sin 45^\circ$

$= \frac{1}{2} \times \frac{1}{\sqrt{2}} - \frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}}$

$= \frac{1 - \sqrt{3}}{2\sqrt{2}} \quad \checkmark$

$= \frac{\sqrt{2} - \sqrt{6}}{4}$

c)  $(1, 4)(5, 2)$   
 $-1: 3$

pt =  $\left(\frac{3 + -5}{2}, \frac{12 + -2}{2}\right)$

$= (-1, 5) \quad \checkmark \checkmark$

d)  $\frac{d}{dx} (4/\sqrt{x}) = -2x^{-3/2}$

$= -2/\sqrt{x^3} \quad \checkmark$

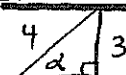
ii.  $d/dx (x^2 - 1)^{10}$

$= 20x(x^2 - 1)^9 \quad \checkmark \checkmark$

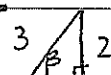
iii.  $\frac{dy}{dx} = \frac{(x+1)(1) - (x)(1)}{(x+1)^2}$

$= \frac{1}{(x+1)^2} \quad \checkmark \checkmark$

### Question 2

a) 

1st quad



2nd quad

i.  $\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha} = \frac{2 \times 3/\sqrt{7}}{1 - 9/7}$

$= \frac{6/\sqrt{7}}{-2/7}$

$= -3\sqrt{7} \quad \checkmark \checkmark$

ii.  $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \sin \beta \cos \alpha$

$= \frac{3}{4} \times \frac{\sqrt{5}}{3} - \frac{2}{3} \times \frac{\sqrt{7}}{4}$

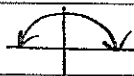
$= \frac{-3\sqrt{5} - 2\sqrt{7}}{12} \quad \checkmark \checkmark$

b)  $\frac{dy}{dx} = vu' + uv'$

$= (x+4)^8 \cdot 2 + 2x[8(x+4)^7]$

$= 2(x+4)^7(x+4 + 8x)$

$= 2(9x+4)(x+4)^7 \quad \checkmark \checkmark$

c)  $\sin 2x = \tan x$  

$2\sin x \cos x = \frac{\sin x}{\cos x} \quad \checkmark$

$2\sin x \cos^2 x - \sin x = 0$

$\sin x (2\cos^2 x - 1) = 0 \quad \checkmark$

$\sin x = 0 \quad \cos^2 x = \frac{1}{2}$

  $\cos x = \pm \frac{1}{\sqrt{2}} \quad \checkmark$

$x = 0, \pi, \pi/4, 3\pi/4$

ie

$x = 0, \pi/4, 3\pi/4, \pi \quad \checkmark$

Question 3

a)  $3x^2 - 4x + 8 = 0$

i.  $\alpha\beta = c/a = 8/3 \quad \checkmark$

$$\begin{aligned} \text{ii. } (\alpha - \beta)^2 &= (\alpha + \beta)^2 - 4\alpha\beta \quad \checkmark \\ &= \left(\frac{4}{3}\right)^2 - 4\left(\frac{8}{3}\right) \\ &= -8\frac{8}{9} \quad \checkmark \end{aligned}$$

c)  $y = ax^3 + bx - \textcircled{1} \quad (1, 0) \quad M_T = 4$

$y' = 3ax^2 + b - \textcircled{2}$

at  $(1, 0) \quad y' = 4$

$$\therefore 0 = a + b \quad \text{and} \quad \leftarrow \begin{array}{l} \leftarrow - \end{array}$$

$4 = 3a + b$

$2a = 4$

$a = 2 \quad \checkmark$

$b = -2 \quad \checkmark$

b) LHS =  $\frac{1 + \cos 2A}{\sin 2A}$

$\sin 2A$

$= \frac{1 + (2\cos^2 A - 1)}{2\sin A \cos A} \quad \checkmark$

$2\sin A \cos A$

$= \frac{2\cos^2 A}{2\sin A \cos A}$

$2\sin A \cos A$

$= \frac{\cos A}{\sin A} \quad \checkmark$

$\sin A$

$= \cot A$

$= \text{RHS}$

ii  $4x - 3y + 2 = 0$

$3y = 4x + 2$

$y = \frac{4}{3}x + \frac{2}{3}$

$M_1 = 4, \quad M_2 = \frac{4}{3}$

$\tan \alpha = \left| \frac{M_1 - M_2}{1 + M_1 M_2} \right| \quad \checkmark$

$= \left| \frac{4 - \frac{4}{3}}{1 + 4 \times \frac{4}{3}} \right|$

$\alpha = 22^\circ 50' \quad \checkmark$

ii.  $\cot 15^\circ = \frac{1 + \cos 30^\circ}{\sin 30^\circ}$

$\sin 30^\circ$

$= \frac{1 + \frac{\sqrt{3}}{2}}{\frac{1}{2}}$

$\frac{1}{2}$

$= 2 + \sqrt{3} \quad \checkmark$

## Question 4.

$$a) i. y' = -1(x^2+1)^{-2} \cdot 2x = \frac{1+2t+t^2}{1-t^2}$$

$$= \frac{-2x}{(x^2+1)^2} \quad \checkmark$$

$$= \frac{(1+t)^2}{(1-t)(1+t)} \quad t \neq \pm 1$$

$$ii. \text{ When } y = 1/2 \text{ and } x < 0$$

$$\frac{1}{2} = \frac{1}{1+x^2} \text{ then } x = -1$$

$$= \frac{1+t}{1-t} \quad \checkmark$$

$$\text{then } M_T = 1/2 \quad \checkmark$$

$$M_N = -2 \quad \checkmark$$

$$ii. \text{ Now } \sec \theta + \tan \theta = 2$$

$$\text{point } (-1, \frac{1}{2}) \text{ and } M_N = -2$$

$$\frac{1+t}{1-t} = 2$$

$$y - \frac{1}{2} = -2(x+1)$$

$$1+t = 2 - 2t$$

$$2y - 1 = -4(x+1)$$

$$3t = 1$$

$$2y - 1 = -4x - 4 \quad \checkmark$$

$$t = 1/3 \quad \checkmark$$

$$4x + 2y + 3 = 0$$

$$\therefore \tan \theta/2 = 1/3$$

$$\theta/2 = 18^\circ 26'$$

$$iii. x\text{-int } y=0 \therefore x = -3/4$$

$$\theta = 36^\circ 52'$$

$$\div 37^\circ \quad \checkmark$$

$$b) \tan \frac{\theta}{2} = t$$

$$\cos \theta = \frac{1-t^2}{1+t^2}$$

$$\tan \theta = \frac{2t}{1-t^2}$$

$$\therefore \sec \theta + \tan \theta$$

$$= \frac{1+t^2}{1-t^2} + \frac{2t}{1-t^2} \quad \checkmark$$

## Question 5

a.  $(\sin A - \cos A)^2 = 1 - \sin 2A$

LHS =  $(\sin A - \cos A)^2$

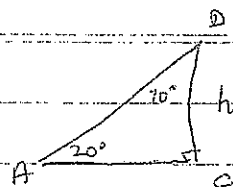
=  $\sin^2 A - 2\sin A \cos A + \cos^2 A$

=  $1 - 2\sin A \cos A$

=  $1 - \sin 2A$

= RHS. ✓

c. i.  $\triangle ADC$



$\tan 70^\circ = \frac{AC}{h}$

$AC = h \tan 70^\circ$  ✓

ii.  $BC = h \tan 80^\circ$  ✓

b)  $\cos \theta - \sqrt{3} \sin \theta$

$R = \sqrt{1^2 + (\sqrt{3})^2}$

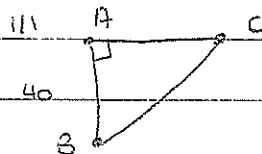
=  $\sqrt{4}$

= 2 ✓

$\tan \alpha = \frac{\sqrt{3}}{1}$

$\alpha = \frac{\pi}{3}$  ✓

$\therefore \cos \theta - \sqrt{3} \sin \theta = 2 \cos \left( \theta + \frac{\pi}{3} \right)$



$BC^2 = 40^2 + AC^2$

$h^2 \tan^2 80^\circ = 1600 + h^2 \tan^2 70^\circ$

$h^2 (\tan^2 80^\circ - \tan^2 70^\circ) = 1600$  ✓

$h^2 = \frac{1600}{\tan^2 80^\circ - \tan^2 70^\circ}$

$h^2 = 65.0015 \dots$

$h = 8.0623 \dots$

ii. Now  $2 \cos \left( \theta + \frac{\pi}{3} \right) = 1$

$\cos \left( \theta + \frac{\pi}{3} \right) = \frac{1}{2}$  ✓

$\theta + \frac{\pi}{3} = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$

$\theta = 0, \frac{4\pi}{3}, 2\pi$  ✓

$h = 8.0623 \dots$

height = 8 metres ✓

## Question 6

a)  $(K+4)x^2 - 3Kx - 4(K-2) = 0$

$$\frac{2\alpha + 1}{K+4} = \frac{3K}{K+4}$$

$$\alpha = \frac{K-2}{K+4} \quad \text{--- (1)} \quad \checkmark$$

$$\alpha(\alpha+1) = \frac{-4(K-2)}{K+4} \quad \text{--- (2)} \quad \checkmark$$

Sub (1) into (2)

$$\frac{(K-2)^2}{(K+4)^2} + \frac{(K-2)}{K+4} = \frac{-4K+8}{K+4}$$

$$(K-2)^2 + (K-2)(K+4) = -4(K-2)(K+4) \quad \checkmark$$

$$(K-2)^2 = -5(K-2)(K+4)$$

$$K=2 \text{ is a soln}$$

also  $K-2 = -5(K+4)$

$$K-2 = -5K-20$$

$$6K = -18$$

$$K = -3$$

$\therefore K=2 \text{ or } K=-3 \quad \checkmark$

$a^2$  can't be neg definite

as  $a < 0, \Delta < 0$  have no common values of  $k$ .

iii. Now  $a > 0, \Delta < 0$

i.e.  $K^2 + K > 0$

$$K < -1, K > 0$$

with  $\Delta < 0$

$$\begin{array}{c} \leftarrow \text{---} 0 \text{---} \rightarrow \\ -9 \quad -1 \quad 0 \quad 1 \end{array}$$

$\therefore K < -9, K > 1$  is solution for positive definite  $\checkmark \checkmark$

b) i Negative definite

$$a < 0, \Delta < 0 \quad \checkmark$$

ii.  $\Delta = b^2 - 4ac$

$$(2K-6)^2 - 4(K^2+K)(2) < 0 \quad \checkmark$$

$$4K^2 - 24K + 36 - 8K^2 - 8K < 0$$

$$-4K^2 - 32K + 36 < 0$$

$$K^2 + 8K - 9 \neq 0$$

$$(K+9)(K-1) \neq 0$$

$$K < -9, K > 1 \quad \checkmark$$

but  $K^2 + K < 0$

$$K(K+1) < 0$$

$$-1 < K < 0$$

$\checkmark$  with conclusion